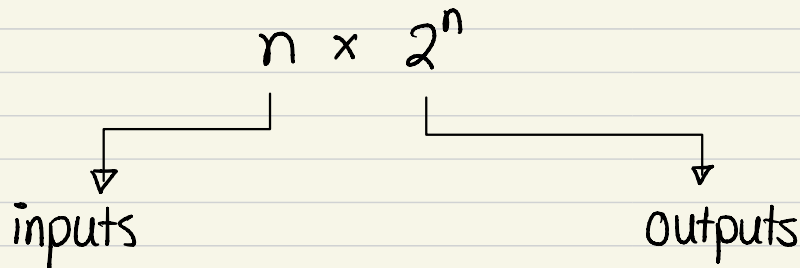


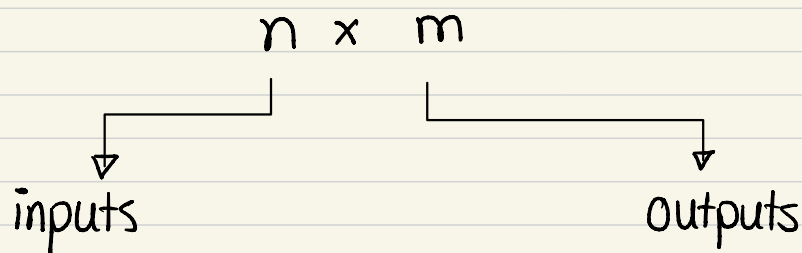
Combinational Circuits

Decoders

Decodes n bits into 2^n unique lines.



Or, if there are unused combinations



where $2^{n-1} < m \leq 2^n$ (if no. of outputs are less than 2^{n-1} , it will require less inputs $(n-1)$ not n)

Example :

3×2^3 : Binary to Octal

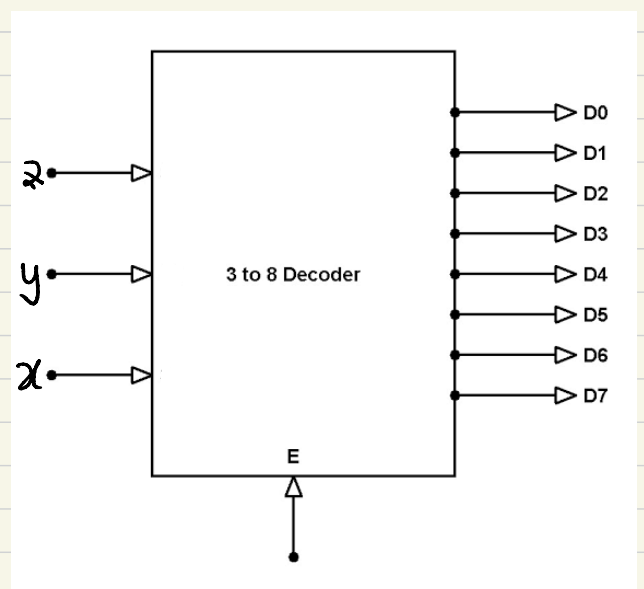
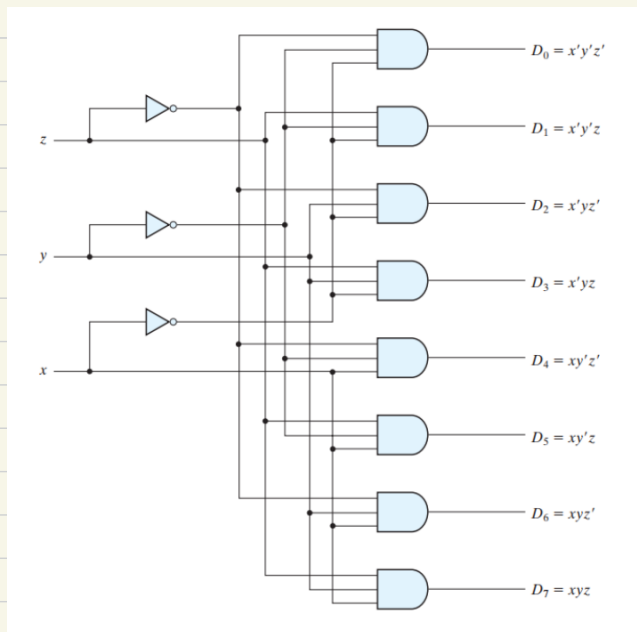
4×10 : BCD to Decimal.

Binary to Octal

All
combination

Inputs			Outputs							
x	y	z	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Per row, only one output active (ON)



Decoder with NAND and ENABLE

NAND - Usually more available and cost efficient.

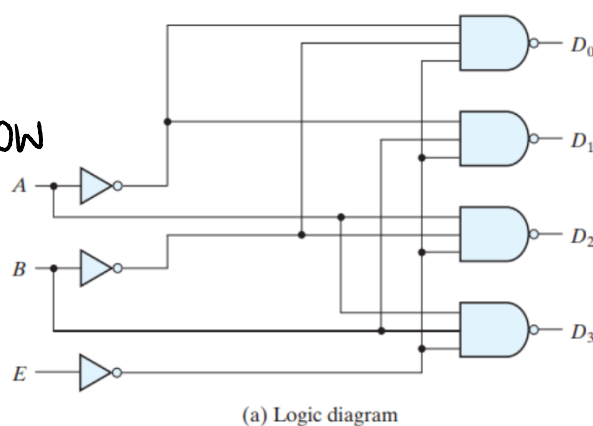
Enable - Decides whether the device will be ON or OFF

modes(2) — $\begin{cases} \text{ACTIVE HIGH, ON when 1} \\ \text{ACTIVE LOW, ON when 0} \end{cases}$

* The mode will be mentioned in the question, if not, you are free to assume as per your convenience.

Here,

E is active LOW



Case I

			Output ACTIVE LOW due to NAND Gate			
E	A	B	D ₀	D ₁	D ₂	D ₃
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

Specialty with "X" in inputs
condense the truth table

Case I: E is 1 (OFF), so inputs won't matter,
Circuit will be INACTIVE: (1111) all OFF

Case II: E is 0 (ON), so, circuit will work.

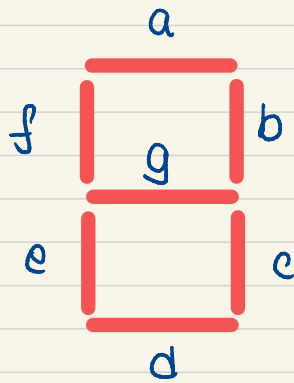
Example.

for input 10 equivalent Octal output will be
ON $\{ (10)_2 = (2)_8 \}$ so D₂ is active (0).

7 segment

* Extra topic for better understanding.

* 7 segment display is an output display device which can show characters and decimal digits.

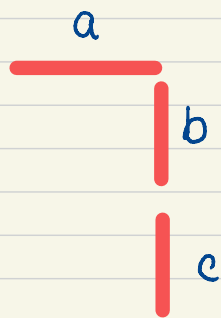


7 segments

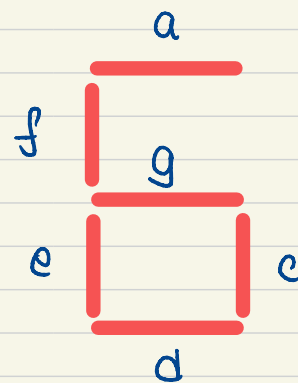
Suppose you want a circuit which takes BCD number as input and show its decimal representation as output in 7 segment display.

Example:

$(0111)_{BCD}$ or $(7)_{10}$



$(0110)_{BCD}$ or $(6)_{10}$



Decimal Digit	Input lines				Output lines							Display pattern
	A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	1	1	1	1	1	1	0	0
1	0	0	0	1	0	1	1	0	0	0	0	1
2	0	0	1	0	1	1	0	1	1	0	1	2
3	0	0	1	1	1	1	1	1	0	0	1	3
4	0	1	0	0	0	1	1	0	0	1	1	4
5	0	1	0	1	1	0	1	1	0	1	1	5
6	0	1	1	0	1	0	1	1	1	1	1	6
7	0	1	1	1	1	1	1	0	0	0	0	7
8	1	0	0	0	1	1	1	1	1	1	1	8
9	1	0	0	1	1	1	1	1	0	1	1	9

Combining Decoders

Construct a 4x16 decoder using 3x8 decoders.

- <https://ade-iitr.vlabs.ac.in/exp/4-to-16/theory.html>
- <https://www.elprocus.com/designing-4-to-16-decoder-using-3-to-8-decoder/>

Input

Output

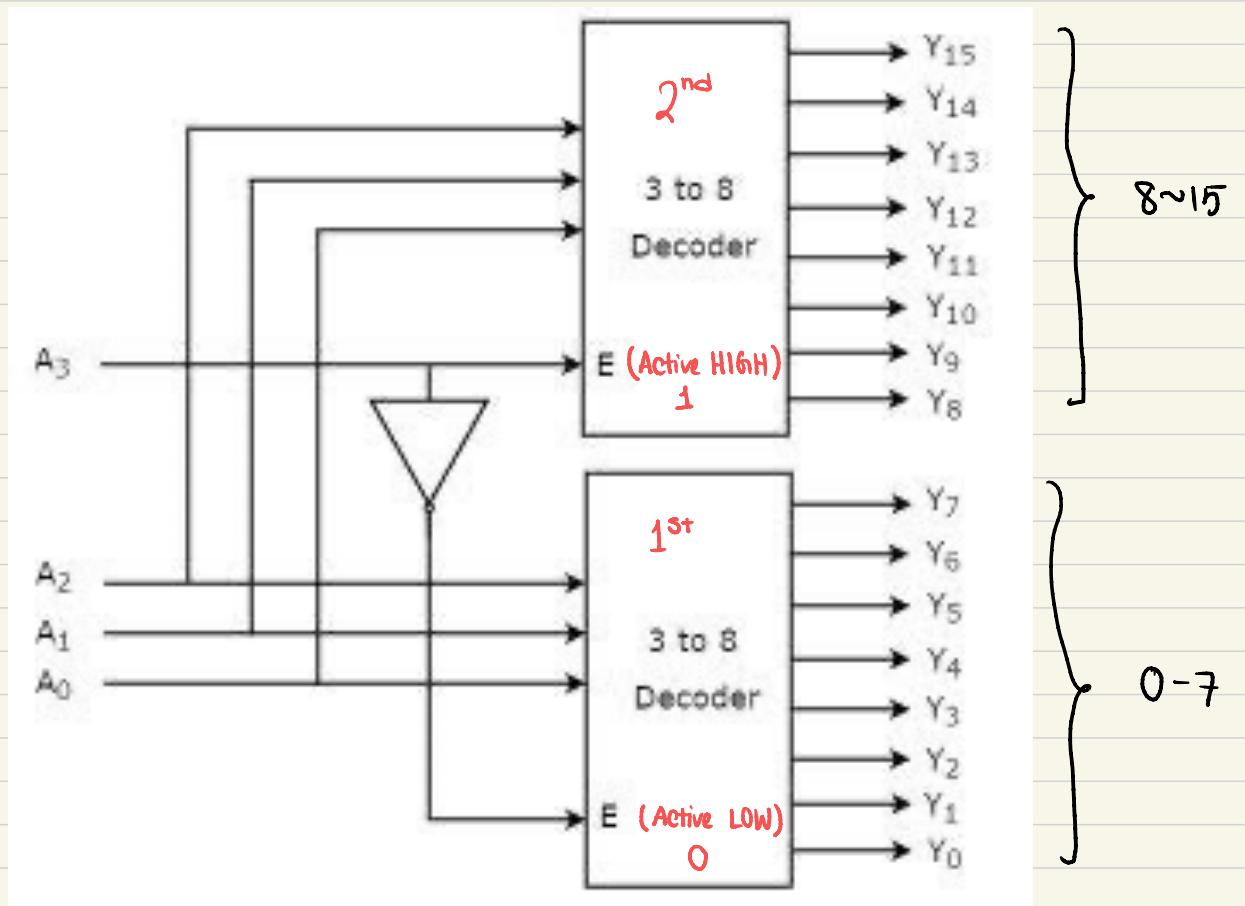
E	A	B	C	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13	Y14	Y15
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

1st decoder
ENABLED,
2nd decoder
DISABLED

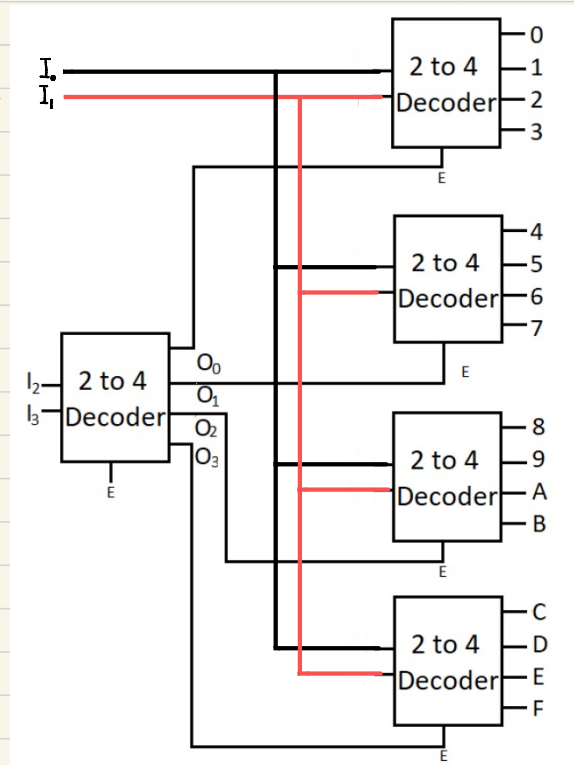
1st decoder
DISABLED,
2nd decoder
ENABLED

ENABLED - ON

DISABLED - OFF



— <https://electronics-fun.com/digital-decoder/>

[illegible]

Construct a 5×32 decoder using 3×8 decoders.

Encoders

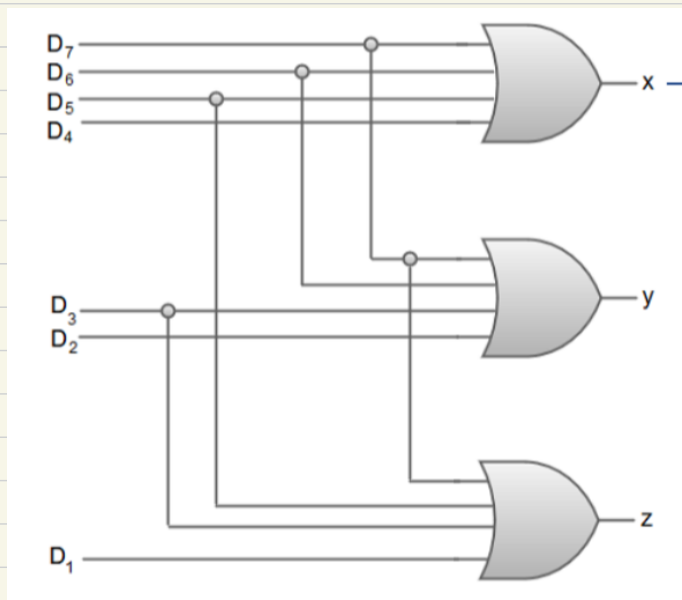
Takes 2^n inputs and provides n outputs.

Inverse of decoders.

Octal to Binary Encoder (3x8 Encoder)

Not all input combinations are used, only 1 active per row (at a time)

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1



$$z = D_1 + D_3 + D_5 + D_7$$

$$y = D_2 + D_3 + D_6 + D_7$$

$$x = D_4 + D_5 + D_6 + D_7$$

Priority Encoder

When there are multiple active/ON inputs at the same time (not defined in simple encoders), what do you expect as output?

From the previous example,

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

If the input is $00\underline{1}00\underline{1}0\underline{1}$ what do you
 $\quad\quad\quad \underline{D_2} \quad \underline{D_5} \quad \underline{D_7}$

expect as output?

D_2 010
 D_5 101
 D_7 111 }

Depends on the
priority of the inputs.

Lets consider the priority to be D_7 (highest) to D_0 (lowest).

Inputs								Outputs		
D_0 (L)	D_1	D_2	D_3	D_4	D_5	D_6	D_7 (H)	x	y	z
1	0	0	0	0	0	0	0	0	0	0
x	1	0	0	0	0	0	0	0	0	1
x	x	1	0	0	0	0	0	0	1	0
x	x	x	1	0	0	0	0	0	1	1
x	x	x	x	1	0	0	0	1	0	0
x	x	x	x	x	1	0	0	1	0	1
x	x	x	x	x	x	1	0	1	1	0
x	x	x	x	x	x	x	1	1	1	1

→ Since D_7 has the highest priority,
If D_7 is 1 (high/on) it won't care about other inputs, so other inputs are don't care (x).

So, $\left. \begin{array}{cccc} 0000 & 000 & 1 \\ \vdots & \vdots & \vdots \\ 1111 & 111 & 1 \end{array} \right\} \begin{array}{l} 111 \\ \text{output} \end{array}$

→ We are done with D_7 , the next input in priority list is D_6 .

D_7 is 0 (otherwise output 111)

D_6 is 1 (2nd highest priority)

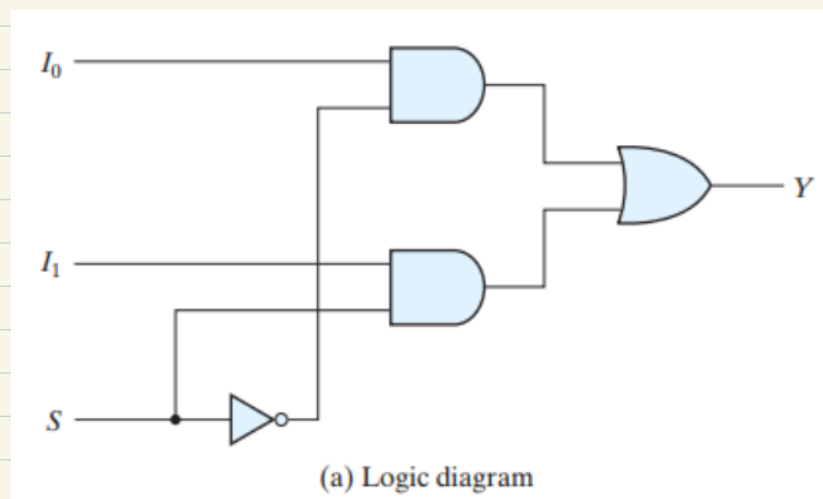
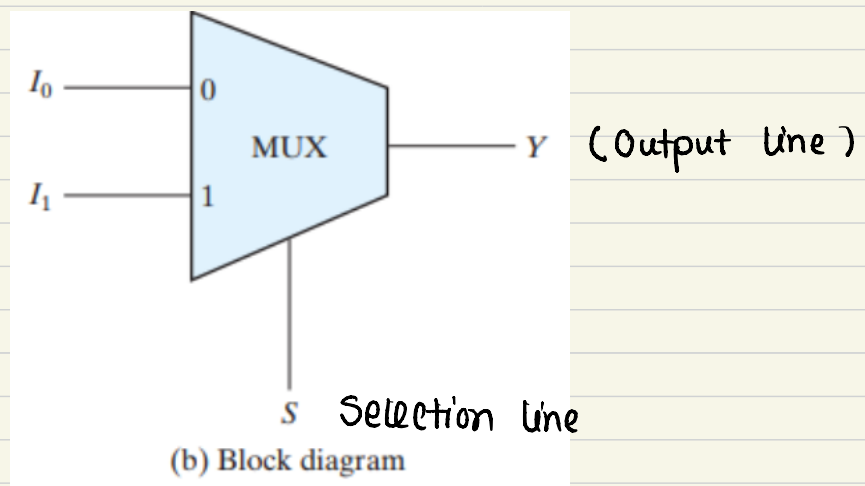
$D_0 \sim D_5$ is irrelevant or don't care (x)

Similarly for other cases ...

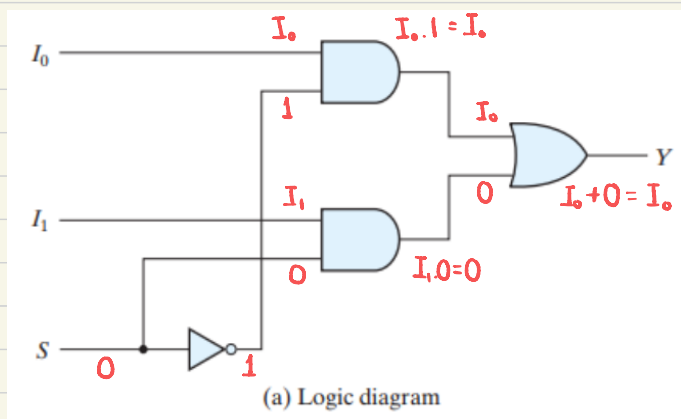
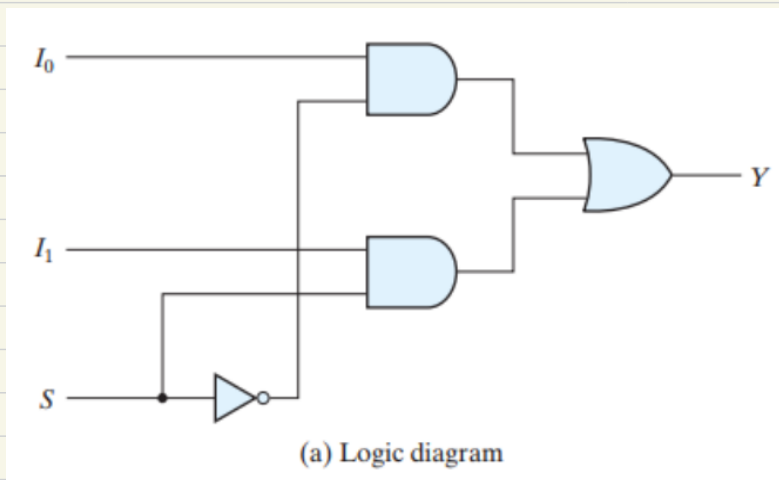
Multiplexer (MUX)

- Data Selector
- **Selects** 1 of many input lines and directs to a single line.
- Input lines : 2^n
Selection Lines : n or $\lceil \log_2(\text{input lines}) \rceil$
Output lines : 1.

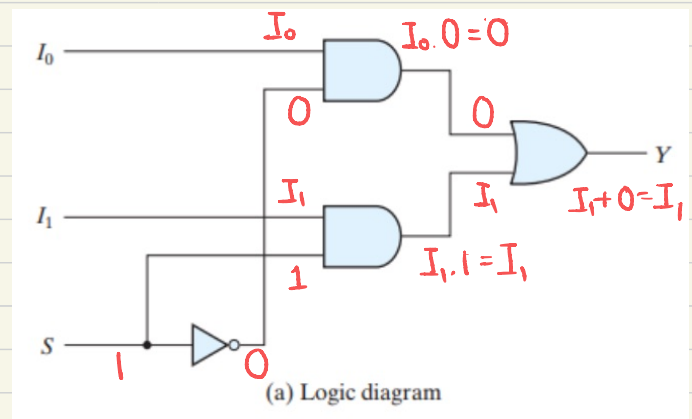
Input



2 to 1 MUX

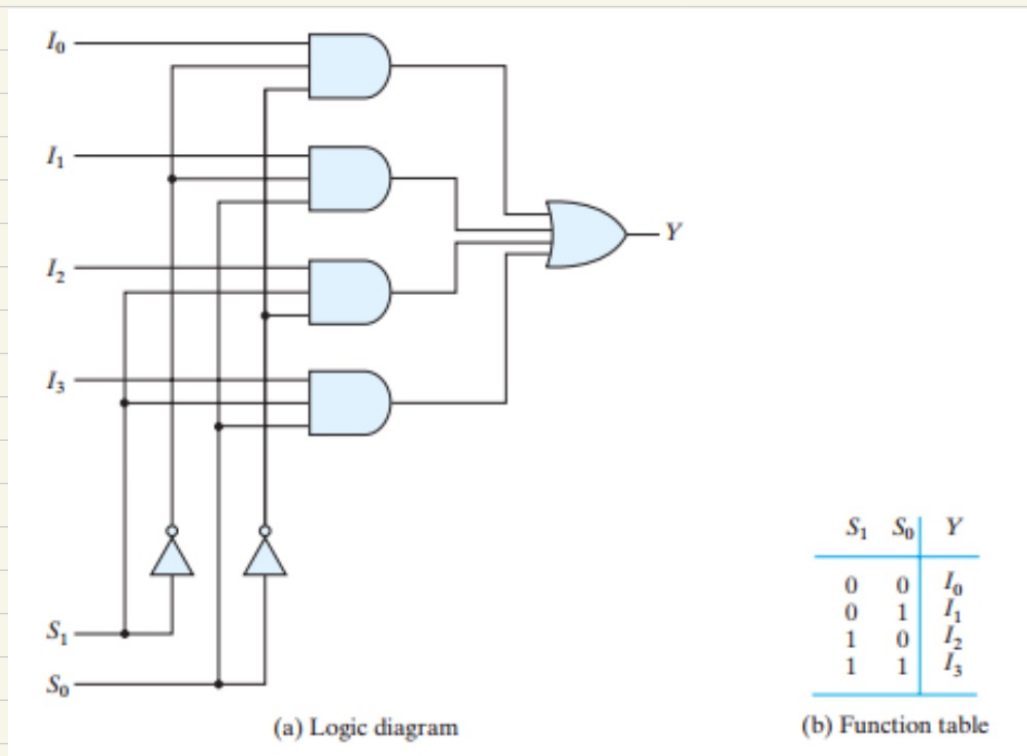


$S=0$, selects $Y = I_0$



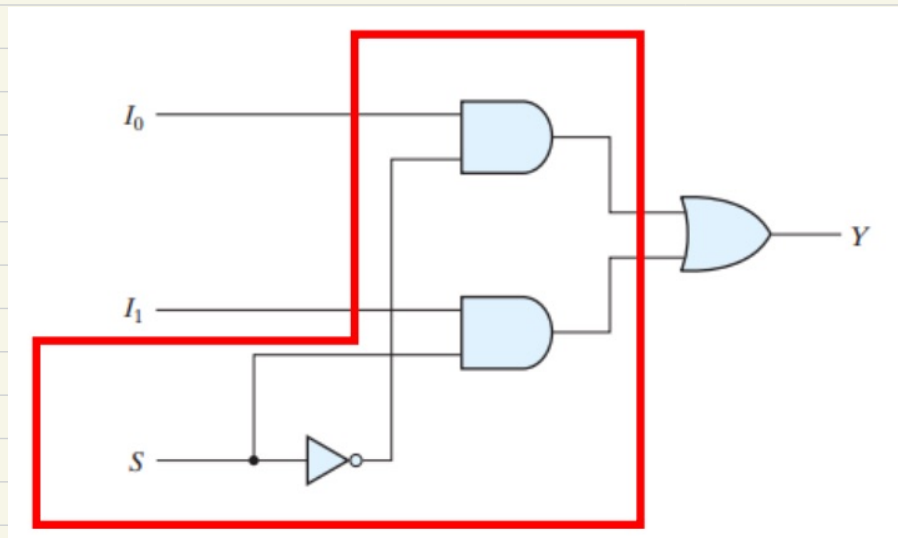
$S=1$, selects $Y = I_1$

4 to 1 MUX

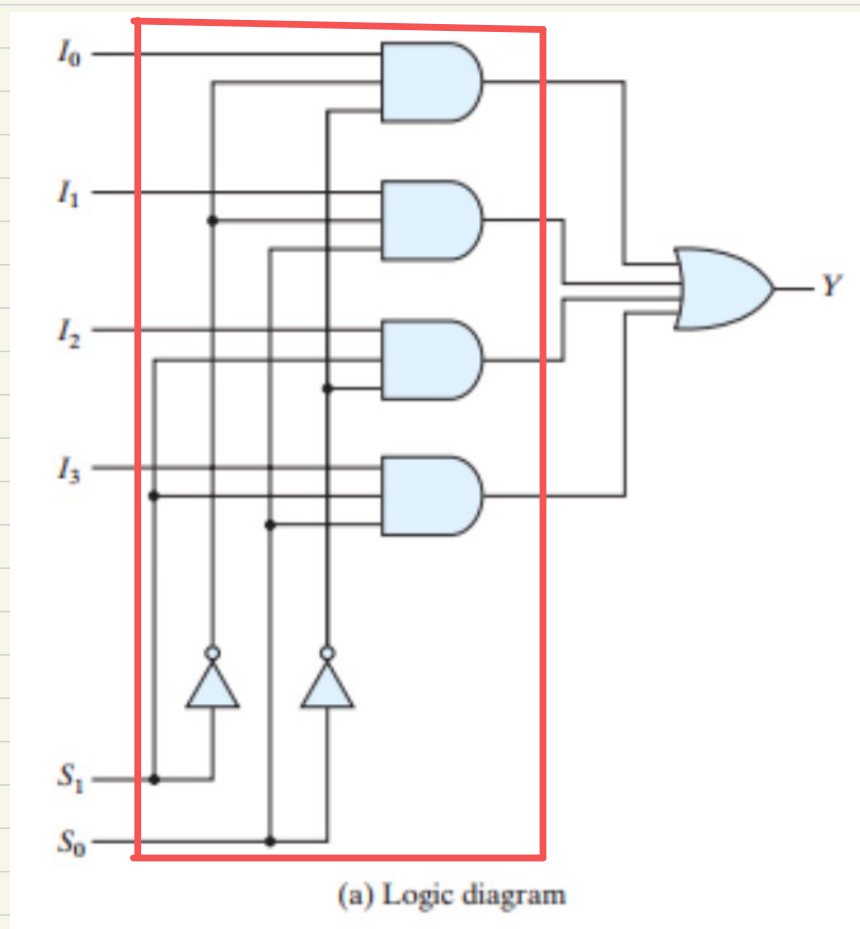


$S_0 S_1$ combination will decide what you'll get as output.

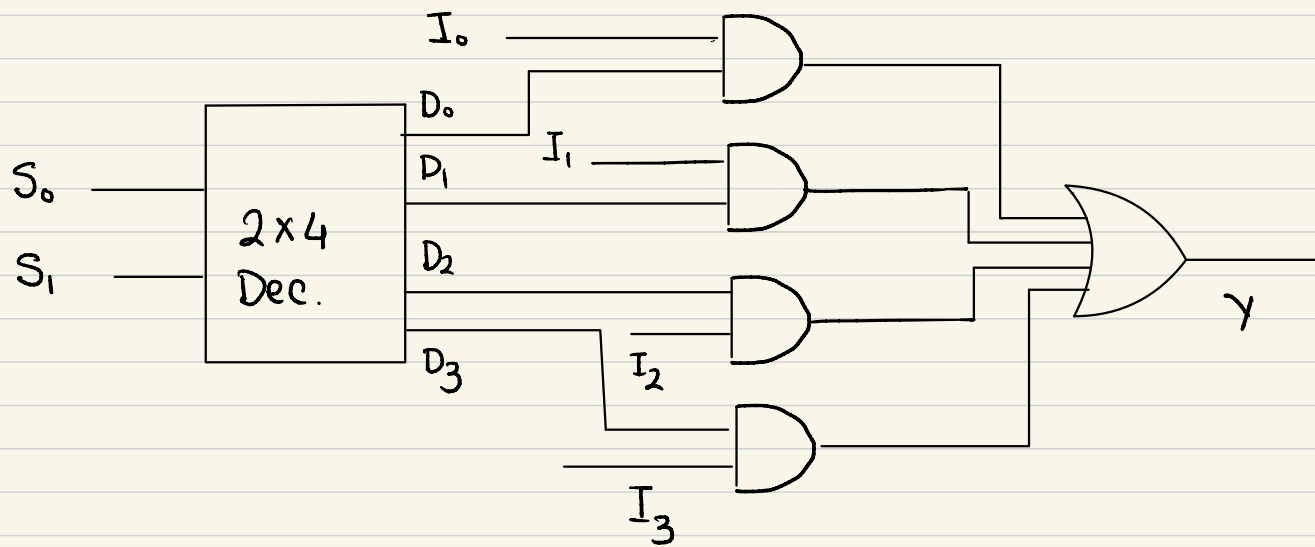
MUX from Decoder



2 to 1 MUX using 1×2^1 decoder



4 to 1 Mux using 2×2^2 decoder.



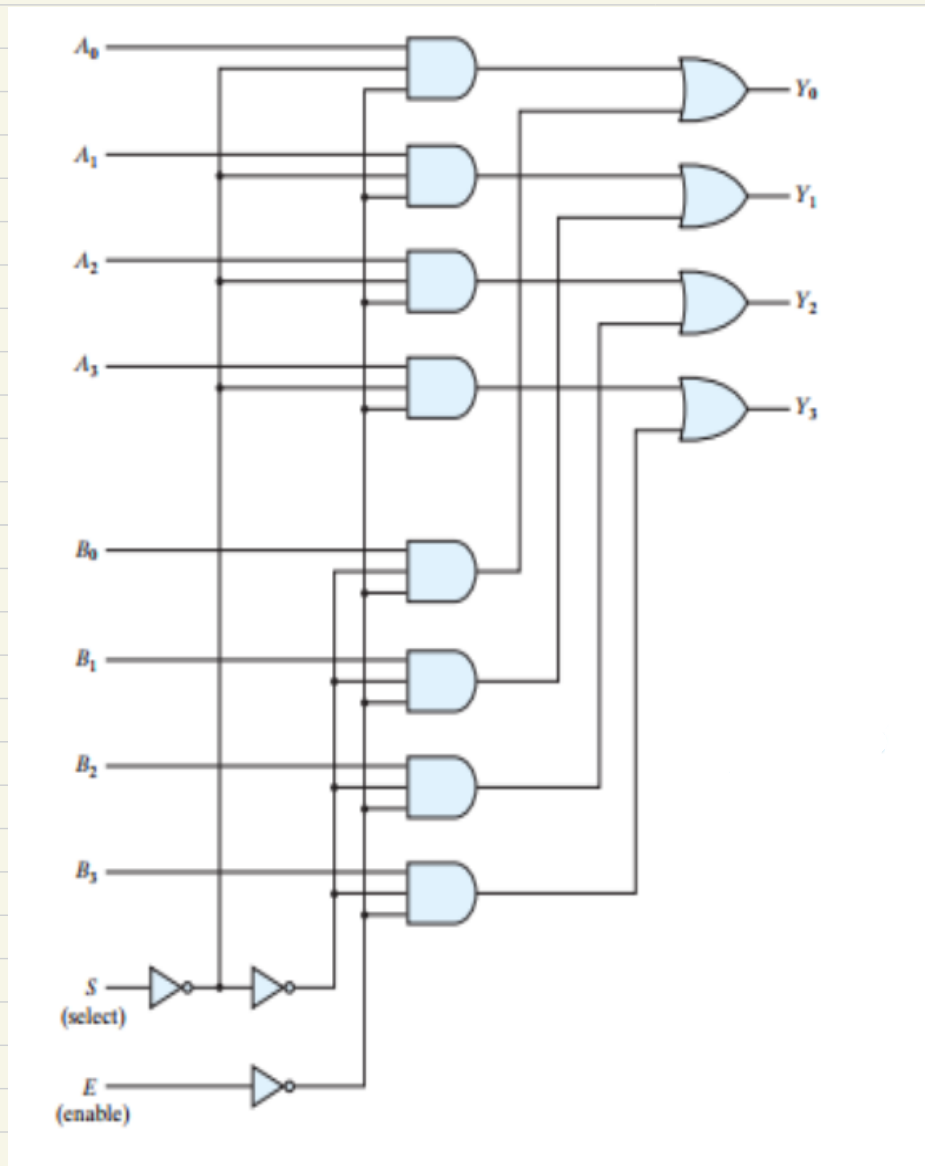
whichever of $D_0 / D_1 / D_2 / D_3$ is 1, that line will pass to Y (output)

Another use of MUX

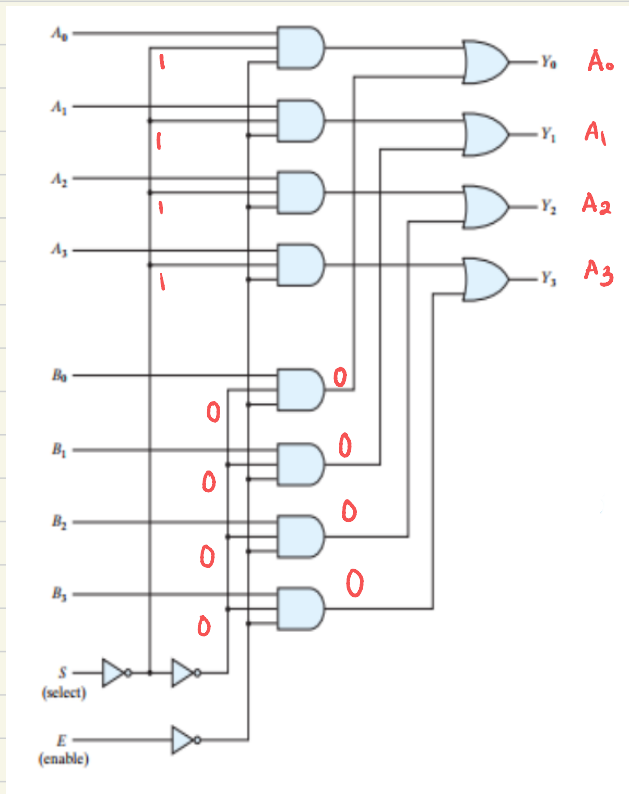
Select multiple lines instead of one.

Example:

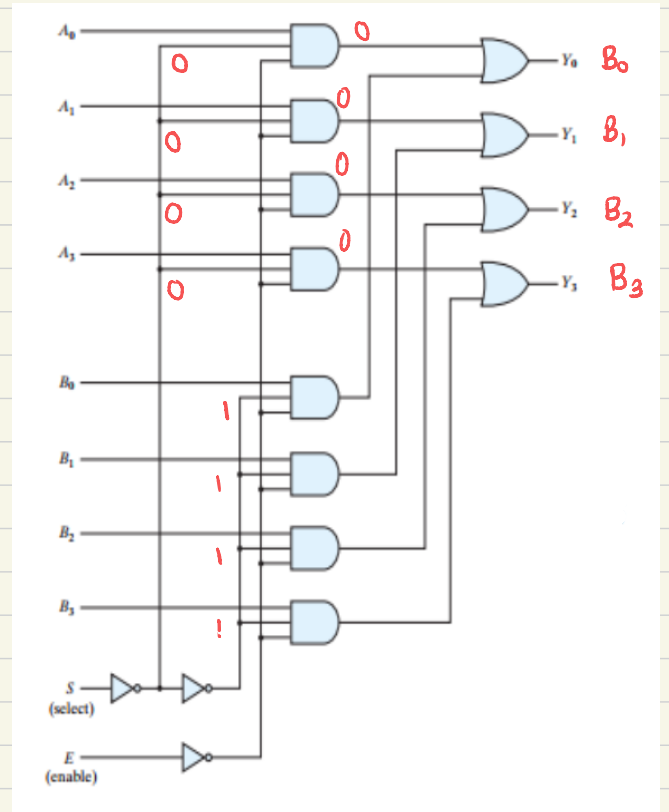
Suppose you have two 4 bit numbers, $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$ and you want to all 4 bits of either A or B as output.



Ignore the Enable line for better understanding.



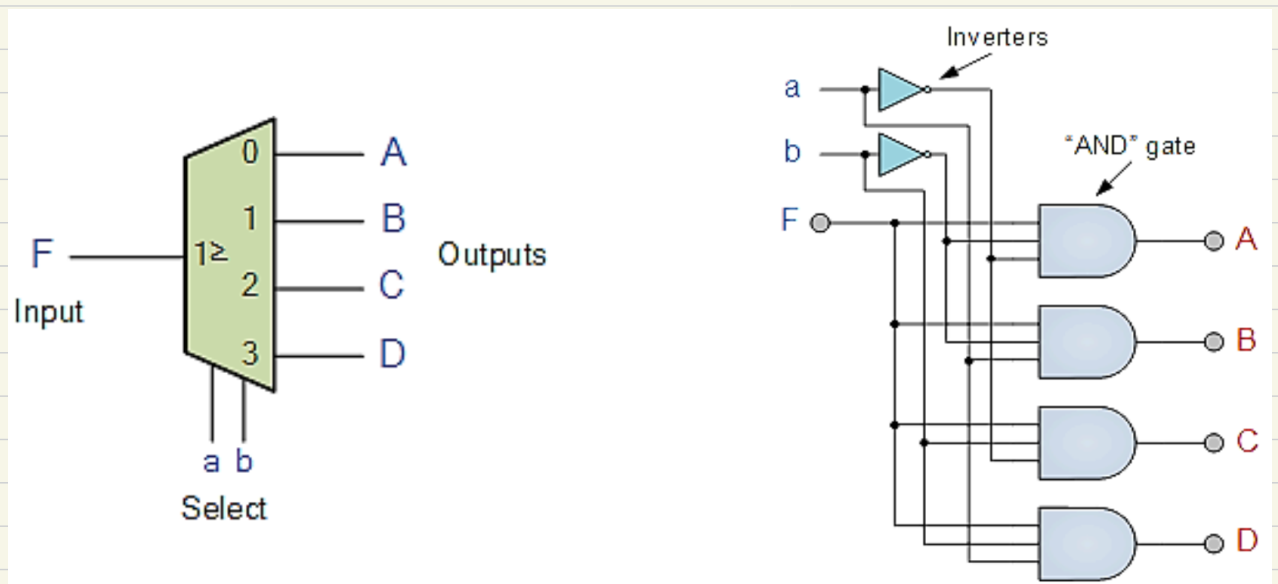
$S = 0$



$S = 1$

Demultiplexer (DEMUX)

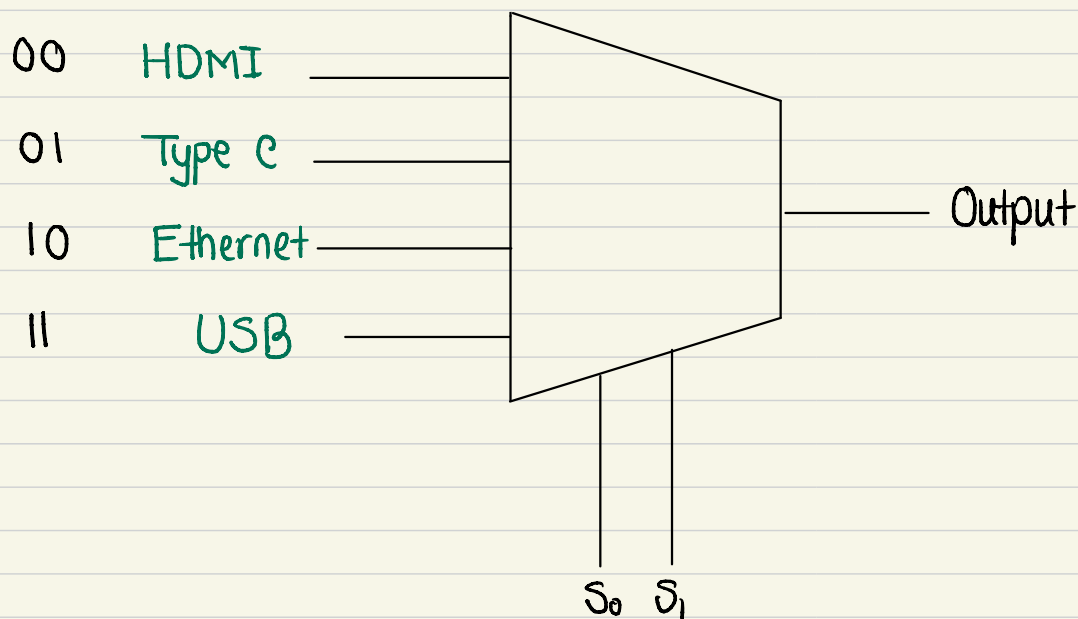
- Data distributor
 - Receives information from a single line and directs it to one of the 2^n possible output lines.
 - Selection controlled by Selection pin.
 - Decoder can be used as DEMUX.
- Input lines : 1
Selection Lines : n or $\lceil \log_2(\text{output lines}) \rceil$
Output lines : 2^n



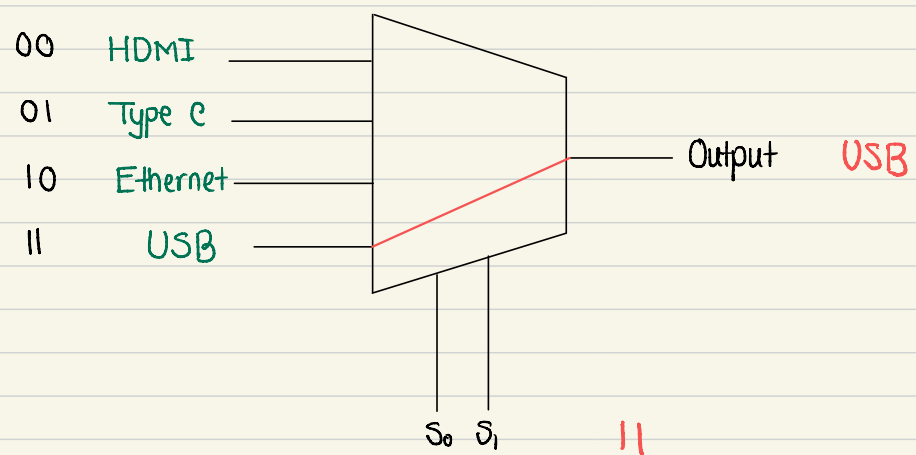
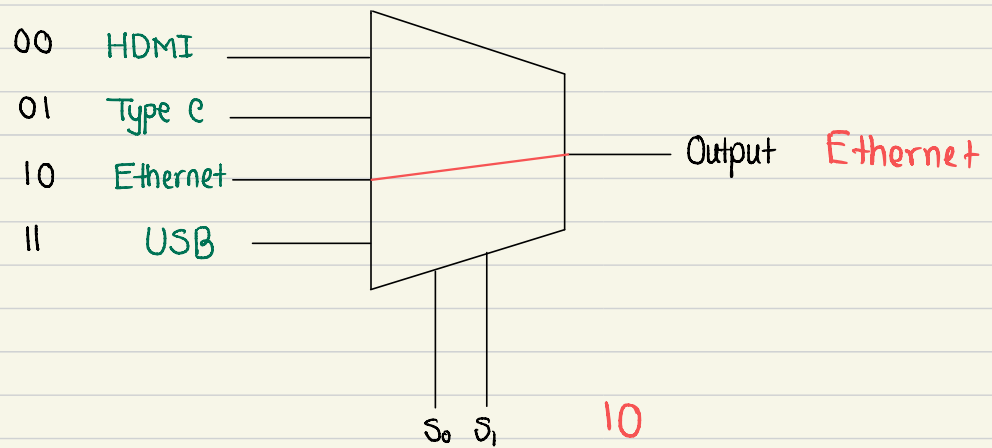
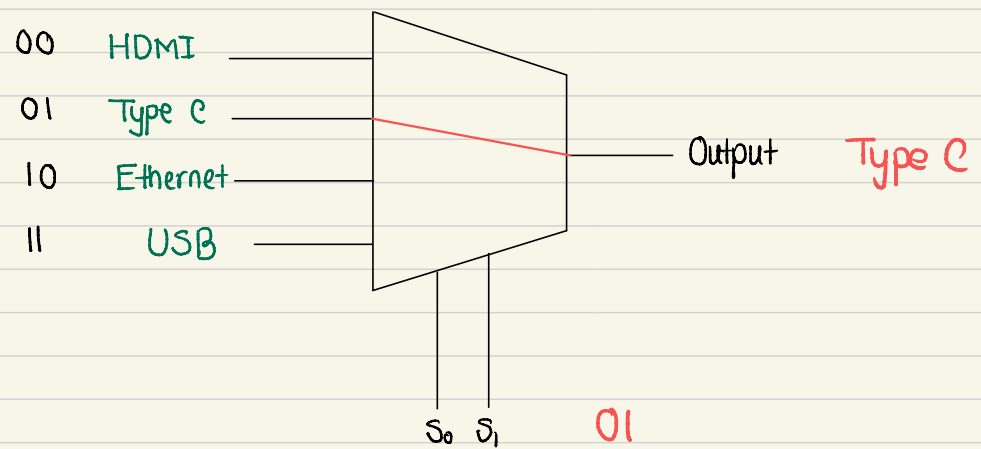
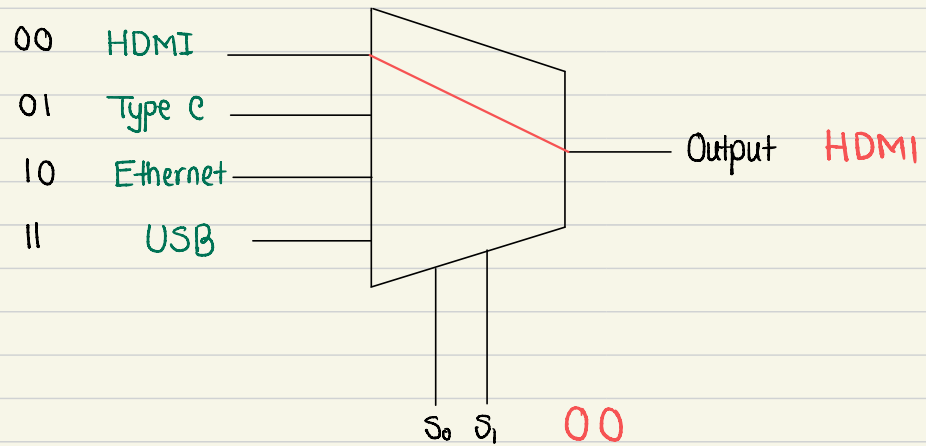
Almost similar to MUX circuit, just OR Gate is missing.

MUX Real Life example

Suppose you have a USB c hub or adapter, which allows to connect multiple types of port or devices to a single main port.

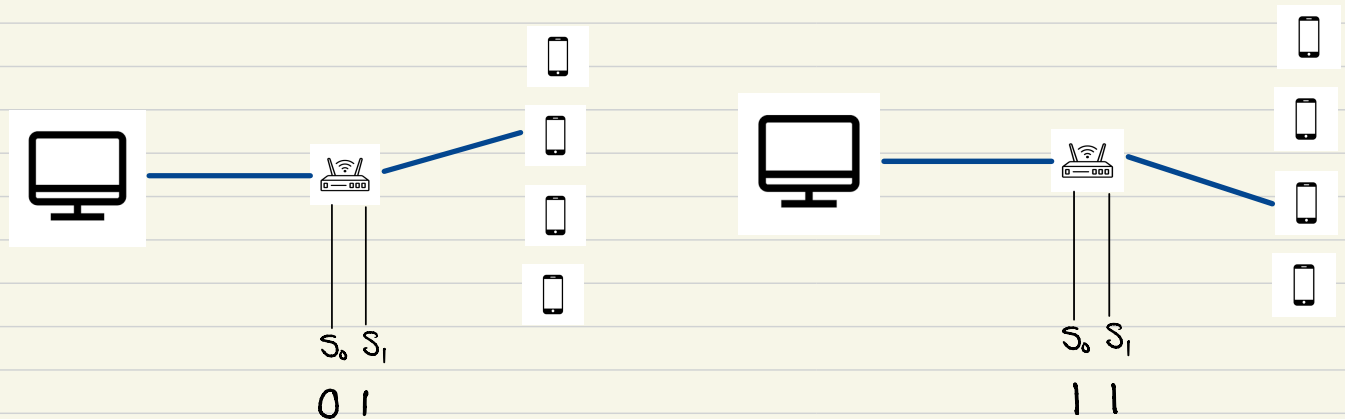
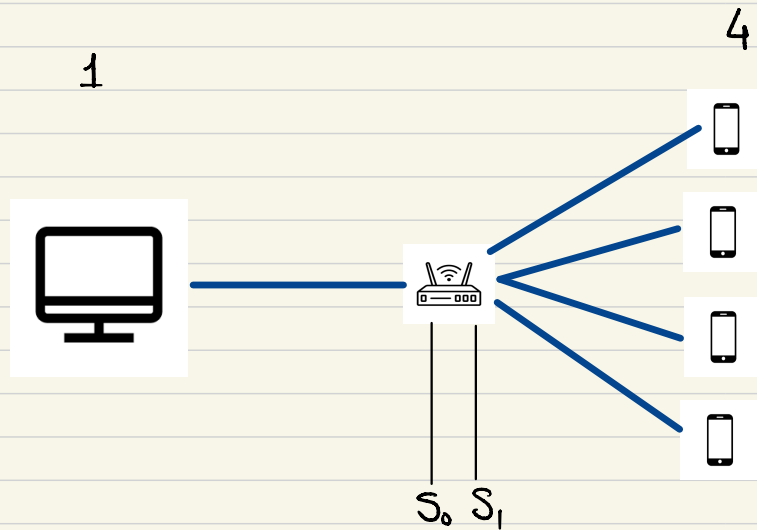


At a time, you can only use one input port which will be directed to the output, the port will be selected using selection lines.



DeMUX

Suppose a PC is connect to a router, and that router is connected to 4 devices, an information is sent from the PC destined for one of the 4 devices, how does the router decide?



info meant for device 1

Info meant for device 2